



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 11



MASSEY FERGUSON FOUR-WHEEL TRACTOR
Model : MF445 Xtra

MASSEY FERGUSON MF445 XTRA FOUR-WHEEL TRACTOR

Test Report No	:	2013 - 11
Date of Test	:	February 13, 14, & 18, 2013
Test Valid Until	:	February 18, 2018
Brand	:	MASSEY FERGUSON
Model	:	MF445 XTRA
Serial No.	:	4454X008006
Type	:	Four-Wheel Drive Tractor
Manufacturer	:	AGCO DO BRASIL COM. E IND. LTDA Av. Guilherme Schell, 10260-Canoas-RS C.G.C. 59.876.003/0001-36, Brazil
Test Requested by	:	ALL CERTIFIED EQUIPMENT TRADING CORPORATION 203 APC Building, 1186 Quezon Ave. Quezon City
Machine Classification	:	Commercial

SUMMARY

The MASSEY FERGUSON MF445 XTRA Four-Wheel tractor was tested to establish its performance. It was tested at the original settings set by the manufacturer.

The Maximum Power developed at the Power-Take-Off (PTO) shaft of the tractor was 58.4 kW at 602 rpm PTO Shaft Speed. The Engine Crankshaft Speed at this point was 2129 rpm with Fuel Consumption Rate of 16.8 L/h. The PTO Shaft Torque at this point was 94.5 kg-m. The Maximum PTO Shaft Torque was 110.6 kg-m which occurred at 410 rpm PTO Shaft Speed.

The Radii of Turning Circle of the tractor when making a right turn were 4900 mm without brake and 4400 mm when brake was applied on the right rear wheel. On the other hand, the Radii of Turning Circle of the tractor when making a left turn were 4875 mm without brake and 4375 mm when brake was applied on the left rear wheel.

The Radii of Turning Area of the tractor when making a right turn were 4940 mm without brake and 4440 mm when brake was applied on the right rear wheel. On the other hand, the Radii of Turning Area of the tractor when making a left turn were 4915 mm without brake and 4415 mm when brake was applied on the left rear wheel.

The Travelling Speeds of the tractor at the engine full throttle speed setting (Pre-set) recommended by the manufacturer/Applicant of test at different gear settings were the following: At “LOW” speed range the travelling Speeds were 2.14 kph, 3.61 kph, 5.89 kph and 3.01 kph at 1st, 2nd, 3rd, and reverse gear settings, respectively, while at “HIGH” speed range the travelling Speeds were 11.2 kph, 18.9 kph, 30.6 kph and 16.2 kph at 1st, 2nd, 3rd, and reverse gear settings, respectively.

SCOPE OF TEST

The tractor was tested to establish its performance characteristics. The test was conducted on February 13, 14, and 18, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna. It was evaluated on the following items:

1. Suitability of operator's manual;
2. Verification of Specifications
3. Laboratory tests
 - a. PTO load performance test;
 - b. Determination of Turning Circle
 - c. Determination of Turning Area
 - d. Determination of Travelling Speeds

METHOD OF TEST

The tractor was tested based on Philippine Agricultural Engineering Standards, PAES 119:2001 Agricultural Machinery – Four-Wheel Tractor – Methods of Test.

DESCRIPTION OF THE TRACTOR

The MASSEY FERGUSON MF445 XTRA Four-Wheel Tractor (**Figure 1**) was a 70.9 kW (95.0 Hp) and had a Rated PTO power of 62.7 kW (84.0 Hp) four-wheel drive tractor as stated in the operator's manual.

The tractor used a four-stroke, water-cooled, turbocharged, four cylinder diesel engine. Fuel feed system was force-feed type using a circulation pump. The fuel tank (**Figure 2**) was located underneath the tractor's steps. It utilized a dry dual type air cleaner (**Figure 3**). Lubrication system was force-feed lubricating type. Cooling was done through circulation of water from cylinder jacket to the radiator using a circulation pump. It was started by an electric starting system. A vertical muffler was used in the exhaust system.

The tractor's transmission system had six forward speeds and two reverse speeds, which were grouped into two ranges. The speed ranges were "LOW" and "HIGH", while the gear settings were 1st, 2nd, 3rd, and reverse. The Main gear was fully synchronized while the range gear could only be shifted when the tractor is completely stopped. The range and main gear settings could be set through a lever at the right side of operator's seat. A parking brake was also provided and can be actuated through a lever at the left side of the operator's seat.

The tractor's direction was being controlled through a hydrostatic power steering system with the steering wheel (**Figure 4**) located in front of the driver's seat. The brakes of the rear wheels were independent of each wheel, however, they could be locked-on through the pedals. A differential lock could be set by foot pedals located on the tractor's flooring just underneath the operator's seat. The size of the rear wheel tires were 46.7 cm x 86.4 cm (18.4 in x 34.0 in) and the front wheel tires were 37.8 cm x 61.0 cm (14.9 in x 24.0 in). Both have spikes.

The tractor had three-point hitch linkage (**Figure 5**) for mounted implements. It had a swinging drawbar (**Figure 6a**) for pulling a trailer or any trailing implement.

The Power-Take-Off shaft (**Figure 6b**) of the tractor was located at the rear of the tractor and was set at 540 rpm. The PTO shaft could be engaged/disengaged through a lever located at the left side of the operator seat with two stage clutch.

The tractor had two headlights, two front auxiliary lights, two brake lights, four turning lights and one working light.

The operator's seat (**Figure 7**) was adjustable for operator's comfort. It was provided with protective seat belt.

The tractor controls (**Figure 8**) were located in front of and beside the operator's seat and included the following: steering wheel, clutch pedal, hand throttle lever, ignition key, main gear shift lever, range gear shift lever, differential lock pedals, brake pedals, foot throttle pedal, PTO controls, hydraulic lift draft/position control lever and auxiliary hydraulic control lever. The panel board (**Figure 8**) which was also located in front of the operator's seat contained the following: tachometer, fuel level indicator, hour meter, battery charge warning light, main warning light, brake fluid level warning light and parking brake application warning light, engine oil pressure warning light, transmission oil pressure warning light, and air filter blockage warning light. **Table 1** shows the detailed specifications of the tractor.



Figure 1. MASSEY FERGUSON MF445 XTRA Four-Wheel Tractor



Figure 2. Fuel tank



Figure 3. Air Cleaner



Figure 4. Steering wheel

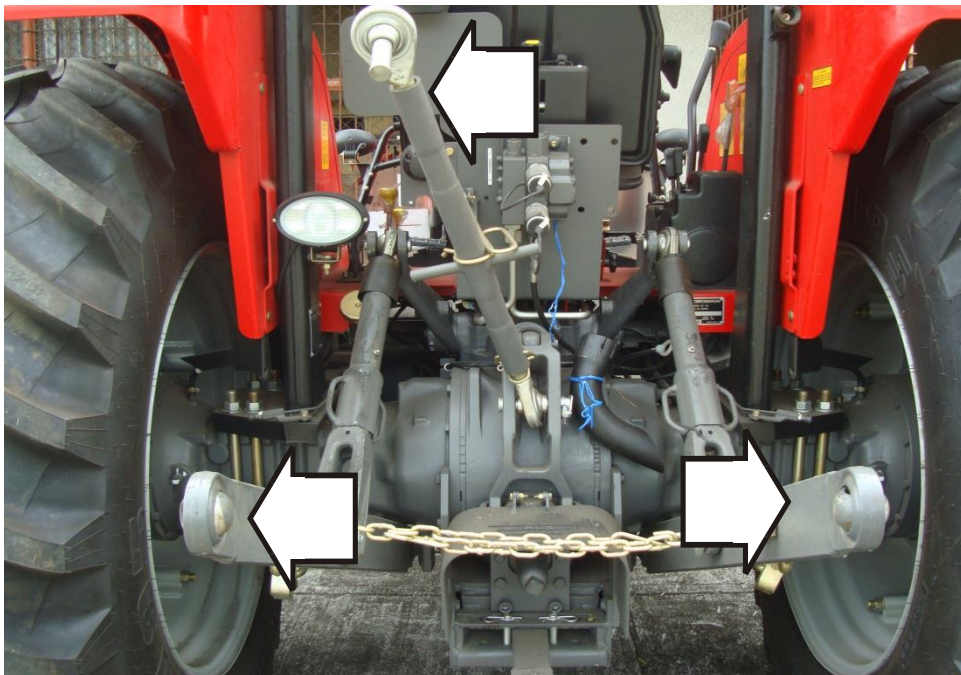


Figure 5. Three-Point Hitch

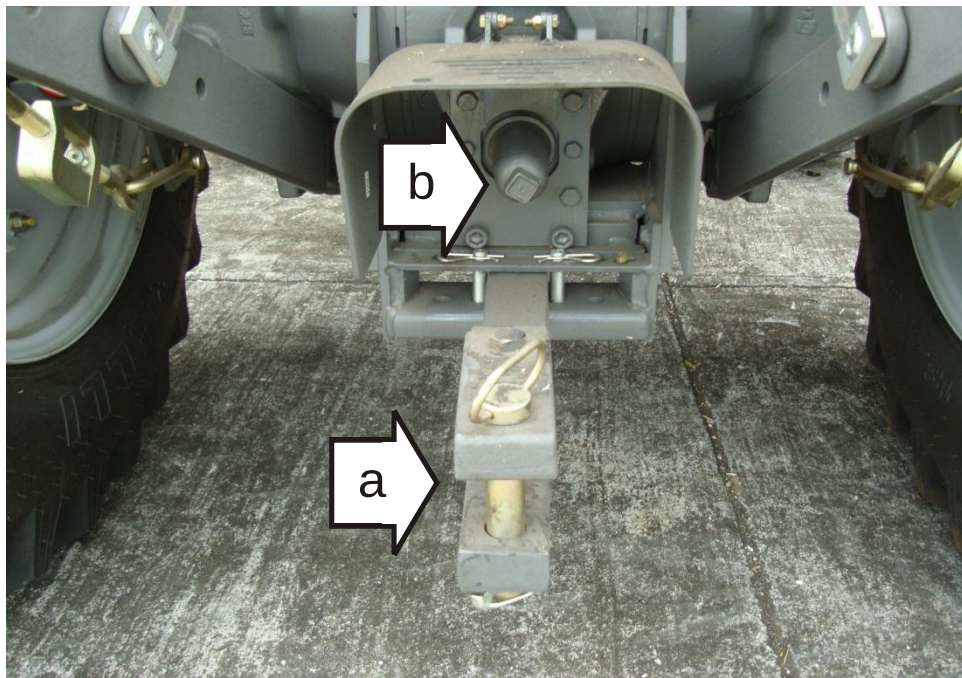


Figure 6. (a) Swinging drawbar
(b) PTO output shaft



Figure 7. Operator's seat



Figure 8. Tractor's control and panel board

SPECIFICATIONS

Table 1. Detailed specifications of the MASSEY FERGUSON MF445 XTRA Four-Wheel Tractor

1.	Tractor specifications (Based on actual ocular inspection of the tractor and operator's manual submitted)		
1.1	Brand	:	MASSEY FERGUSON
1.2	Model	:	MF445 XTRA
1.3	Serial No.	:	4454X008006
1.4	Make	:	AGCO DO BRASIL
1.5	Type	:	Four-Wheel Drive
1.6	Rated Maximum Power (kW)	:	70.9 (Based on operator's manual)
1.7	Rated Maximum PTO Power (kW)	:	62.7 (Based on Operator's manual)
2.	Dimensions and Weight		
2.1	Overall length (mm) With front ballast	:	4400
2.2	Overall width (mm)	:	2165
2.3	Overall height (mm)	:	2685 (Canopy)
3.	Tractor's engine		
3.1	Brand	:	PERKINS
3.2	Type	:	Four-Stroke Cycle, Water-cooled, Turbocharged, Four Cylinder Diesel Engine
3.3	Model	:	1104A-44T
3.4	Make	:	Brazil
3.5	Serial Number	:	RS60153B505411W
3.6	Rated Output Power (kW)	:	70.9
3.7	Number of Cylinder	:	4

Table 1. Con't

3.7	Bore x stroke (mm)	:	No available data
3.8	Total displacement volume (cc)	:	4100
3.9	Fuel system		
3.9.1	Type of fuel feed	:	Circulation pump
3.9.2	Fuel used	:	Diesel
3.10	Cooling system	:	Water-cooled
3.11	Lubricating system	:	Force-feed type
3.12	Starting system	:	Electric type using battery
3.13	Air cleaner	:	Dry dual type
3.14	Exhaust system	:	Vertical muffler
4.	Power-Take-off		
4.1	Location	:	At the rear of the tractor
4.2	Diameter of PTO shaft end (mm)	:	35.0
4.3	Number of splines	:	6
4.4	Height above ground (mm)	:	635
4.5	Rated Shaft Speed(rpm)	:	540
4.6	Direction of rotation (viewed from the rear of the tractor)	:	Clockwise
4.7	Mode of Operation	:	Lever
5.	Drawbar		
5.1	Type	:	Swinging
5.2	Diameter of hitchpoint (mm)	:	28.0
6.	Tire		
6.1	Front		
6.1.1	Number	:	2
6.1.2	Size (cm)	:	37.8 x 61.0
6.1.3	Track width (mm)	:	1860

Table 1. Con't

6.2	Rear		
6.2.1	Number	:	2
6.2.2	Size	:	46.7 x 86.4
6.2.3	Track width (mm)	:	1640
7.	Wheel base (mm)	:	2330
8.	Ground Clearance (mm)	:	380 (Drawbar bracket)

Table 2. Dimension of linkage geometry of MASSEY FERGUSON MF445 XTRA tractor

(see also **Figure 9**)

Particulars		Data (mm)
1. Length of Lift Arms	A	300
2. Length of Lower Links	B	1070
3. Horizontal distance between two Lower Link Points	u	480
4. Horizontal distance between the two Lift Arm End Points	v	520
5. Length of upper link	S	740 - 985
6. Distance of Lower Link Pivot Points to Lift Rod Pivot Points on Lower Links	D	555
7. Length of Lift Rods	L	650 - 925
8. Height of Lower Hitch Points relative to the rear wheel axis		
a. Low Position	h	220
b. High Position	H	330
9. Height above ground of Lower hitch points when locked in transport position		985
10. Three-point-hitch		Category II

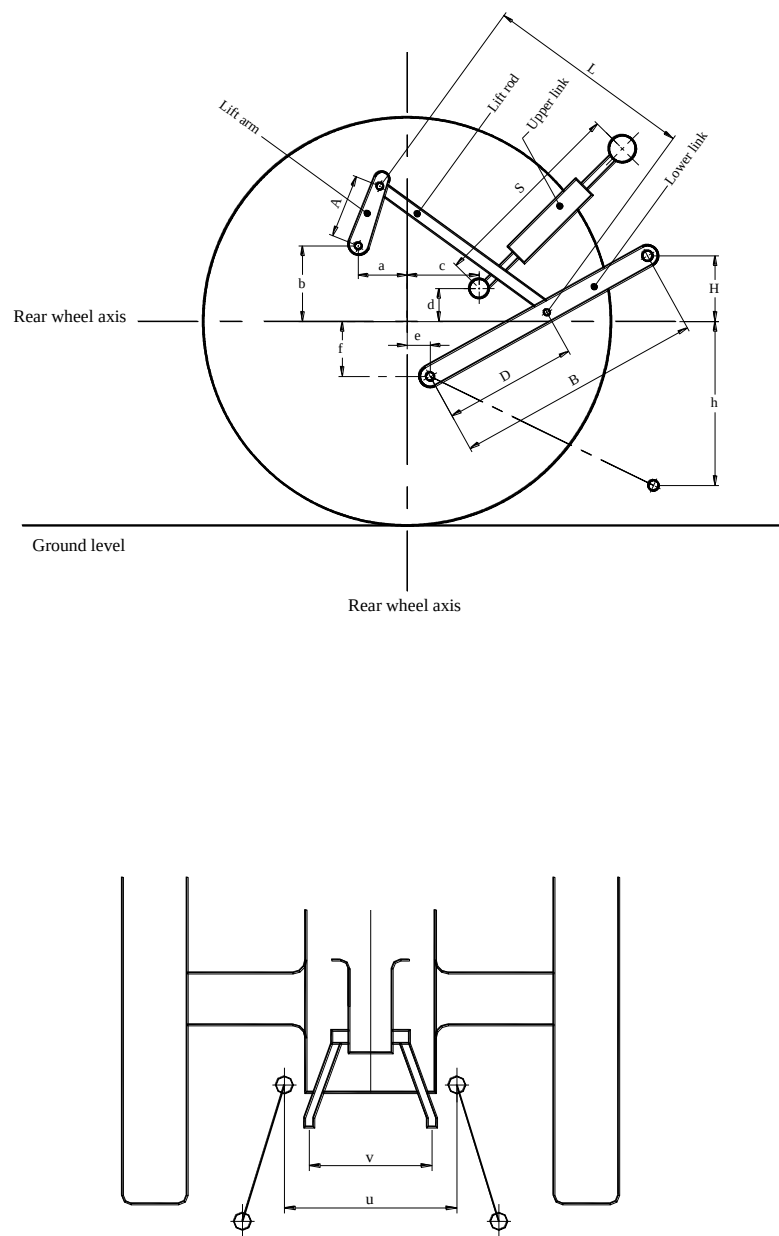


Figure 9. Linkage Geometry

OPERATOR'S MANUAL

The operator's manual submitted with the tractor was presented in English and contained the following: Safety guidelines; Identification and safety decals, Instruments and commands, Preparation for operation, Operation, Maintenance, Specifications, and Accessories.

RESULTS AND DISCUSSION

The MASSEY FERGUSON MF445 XTRA Four-Wheel tractor was tested to establish its performance. It was tested at the original settings set by the manufacturer.

A. DETERMINATION OF TURNING CIRCLE AND TURNING AREA

The tractor was tested in a suitable concrete area. It was tested by operating the tractor at the slowest speed possible. The Steering wheel was rotated to the maximum that it can rotate during turning to the left and right directions. The Radii of Turning Circle and Turning Area with and without brakes were measured. The Radius of Turning Circle is the Radius of the smallest circle tangentially described by the median plane of the outermost wheel, while the Radius of Turning Area is the radius of the smallest circle described by the outermost point of the tractor. Results of the test are shown in **Tables 3** and **4**.

TURNING CIRCLE AND TURNING AREA

Tractor on Test : MASSEY FERGUSON
 Model : MF445 XTRA
 Date of Test : February 18, 2013

Table 3. The Radius of Turning Circle of the tractor

	RIGHT TURN	LEFT TURN
WITHOUT BRAKES	4900	4875
WITH BRAKES	4400	4375

Table 4. The Radius of Turning Area of the tractor

	RIGHT TURN	LEFT TURN
WITHOUT BRAKES	4940	4915
WITH BRAKES	4440	4415

B. DETERMINATION OF TRAVELLING SPEEDS

The Travelling Speeds of the tractor were measured in actual condition. These were measured on a 50-meter concrete road. The tractor was operated at full throttle speed setting and made to run at different range and speed settings. The time elapsed for the tractor to cross the 50-meter distance was obtained. The third gear setting at range “HIGH”, yielded the fastest Forward Travelling speed of 30.6 kph, while the first gear setting at range “LOW” was slowest at 2.14 kph. For the Reverse Travelling Speed, the speed were 3.01 kph and 16.2 kph for the “LOW” and “HIGH” range settings, respectively. In general, the tractor speeds increased with increased range and gear settings. **Table 5** shows the results of the test.

TRAVELLING SPEEDS

Tractor on Test : MASSEY FERGUSON
 Model : MF445 XTRA
 Date of Test : February 18, 2013

Table 5. The Travelling Speeds of the Tractor in kph at different settings

RANGE GEAR SETTINGS	MAIN GEAR SETTINGS (kph)			
	FIRST	SECOND	THIRD	REVERSE
LOW	2.14	3.61	5.89	3.01
HIGH	11.2	18.9	30.6	16.2

C. PTO PERFORMANCE LOAD TEST

The power at the Power-Take-Off shaft of the tractor was obtained using a Portable Waterbrake Dynamometer. It was tested at full throttle speed setting of the tractor. Data such as dynamometer load, PTO rpm, Engine rpm, fuel consumption were obtained at 20 & 40 rpm speed increments. The Maximum No-Load speed of the engine was 2304 rpm. The Maximum PTO Output Power obtained was 58.4 kW at 602 rpm PTO shaft speed. At this point, the PTO torque was 94.5 kg-m with tractor's Fuel Consumption Rate of 16.8 L/h. No breakdown and other malfunctioning occurred during the test. The test set-up is shown in **Figure 10** and the results of test are shown in **Table 6** and plotted in **Figure 11**.



Figure 10. The Tractor on the PTO test set-up

PTO PERFORMANCE TEST

Tractor on Test : MASSEY FERGUSON
 Model : MF445 XTRA
 Date of Test : February 13-14, 2013

Table 6. Results of PTO load performance test

SHAFT SPEED (rpm)		PTO		FUEL CONSUMPTION (L/h)	SPECIFIC FUEL CONSUMPTION (g/kW-h)
ENGINE	PTO	TORQUE (kg-m)	POWER (kW)		
2304	653	12.6	8.40	7.06	703.0
2208	624	89.1	57.1	17.1	252.2
2129	602	94.5	58.4	16.8	241.8
2058	581	96.6	57.6	16.5	239.9
1990	562	98.8	57.0	16.4	241.4
1923	542	101.1	56.3	16.2	241.5
1828	520	103.8	55.5	15.5	234.6
1765	500	106.3	54.6	15.2	233.3
1624	455	109.1	50.9	14.6	239.9
1458	410	110.6	46.5	12.9	233.3

REMARKS / OBSERVATIONS: No breakdown and any malfunctioning occurred.

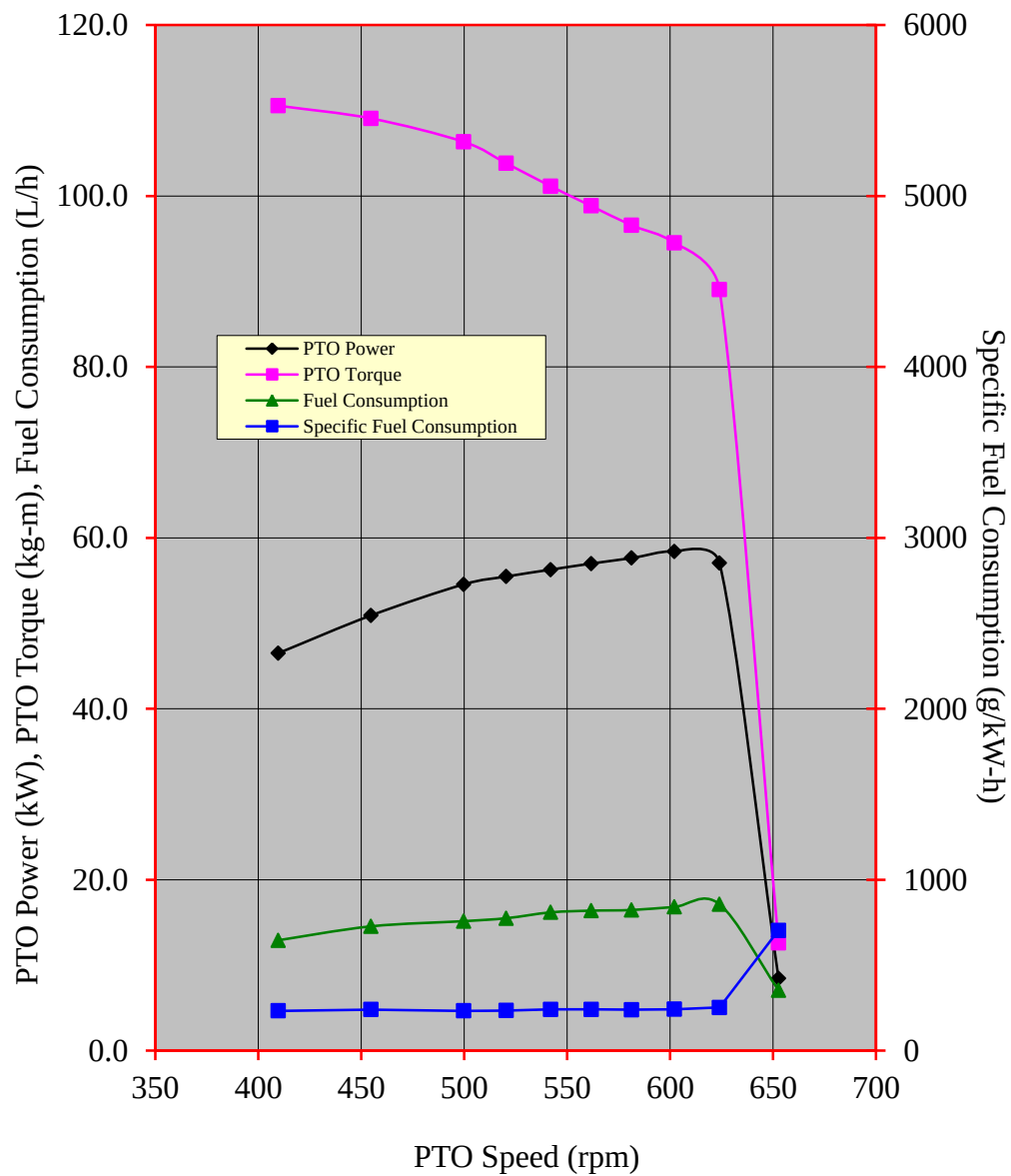


Figure 11. Performance Characteristic Curves of the MASSEY FERGUSON MF445 XTRA Four-Wheel Tractor

Test Engineer:

ROMULO E. EUSEBIO

Checked by:

DARWIN C. ARANGUREN

Approved for release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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